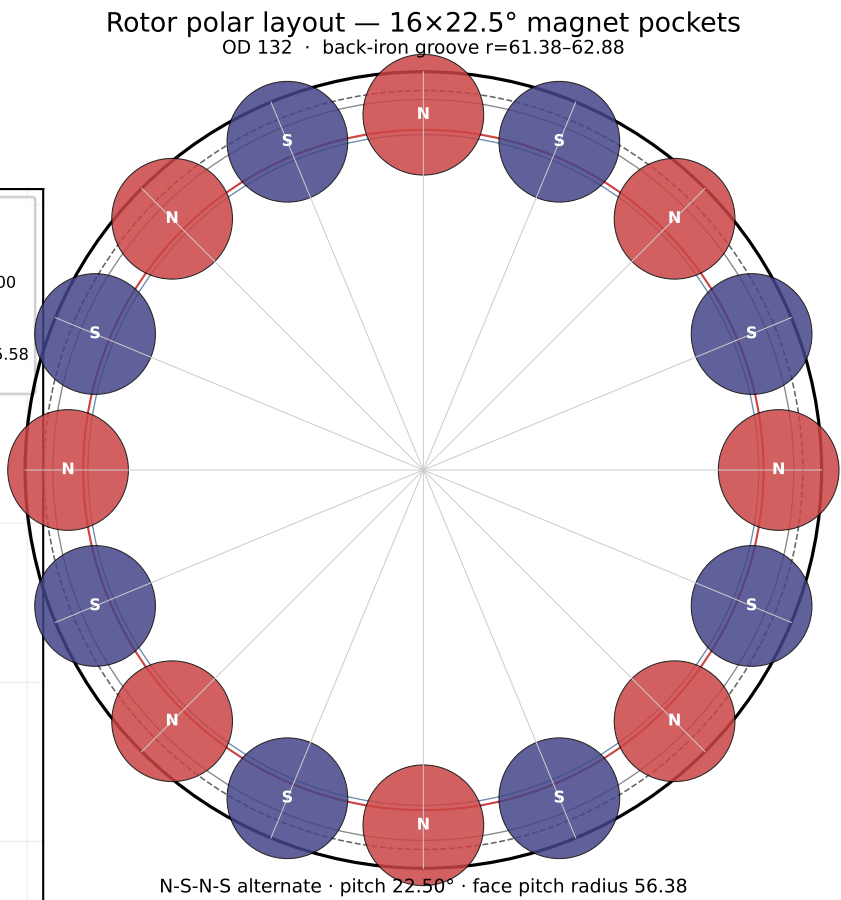
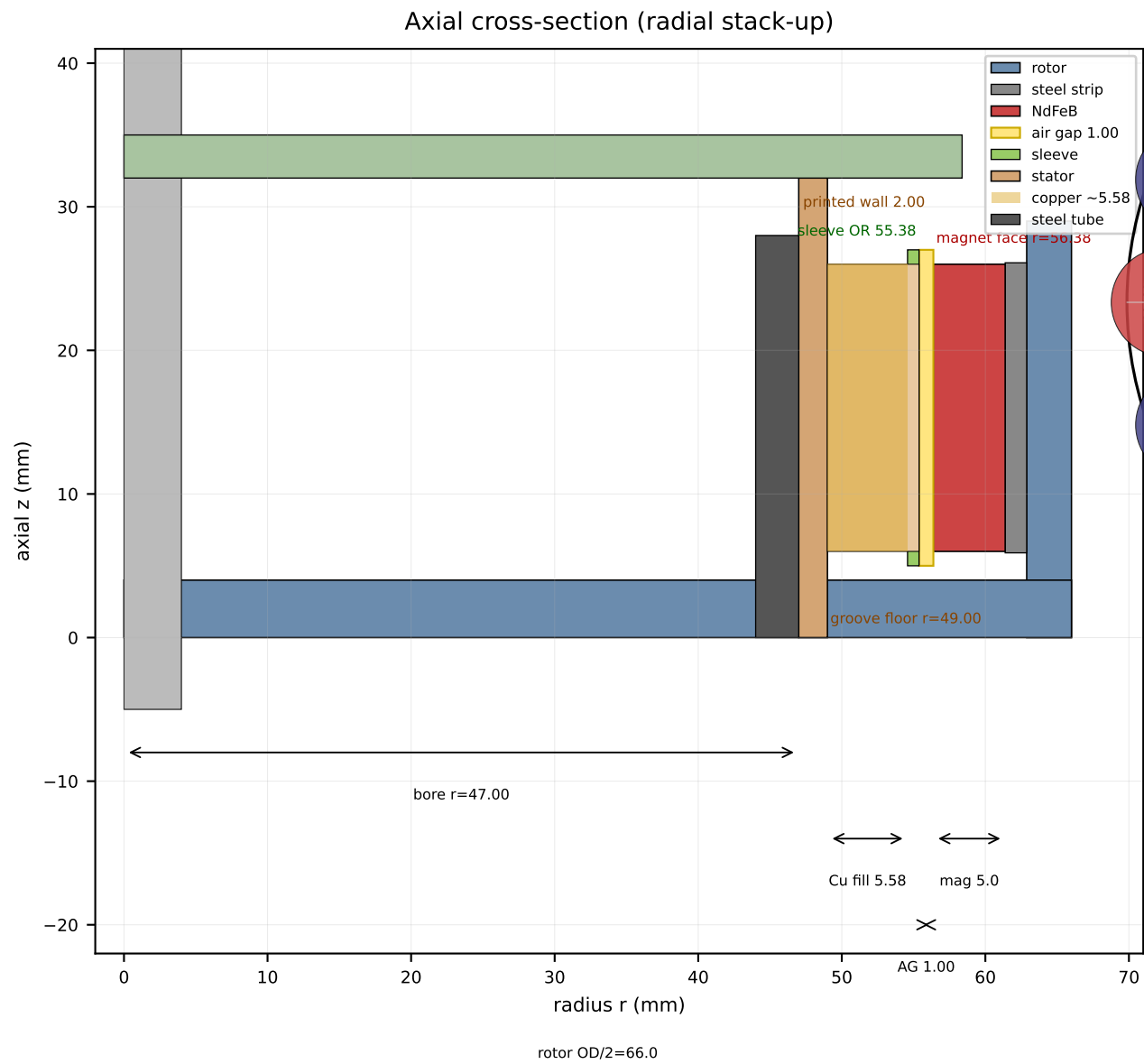


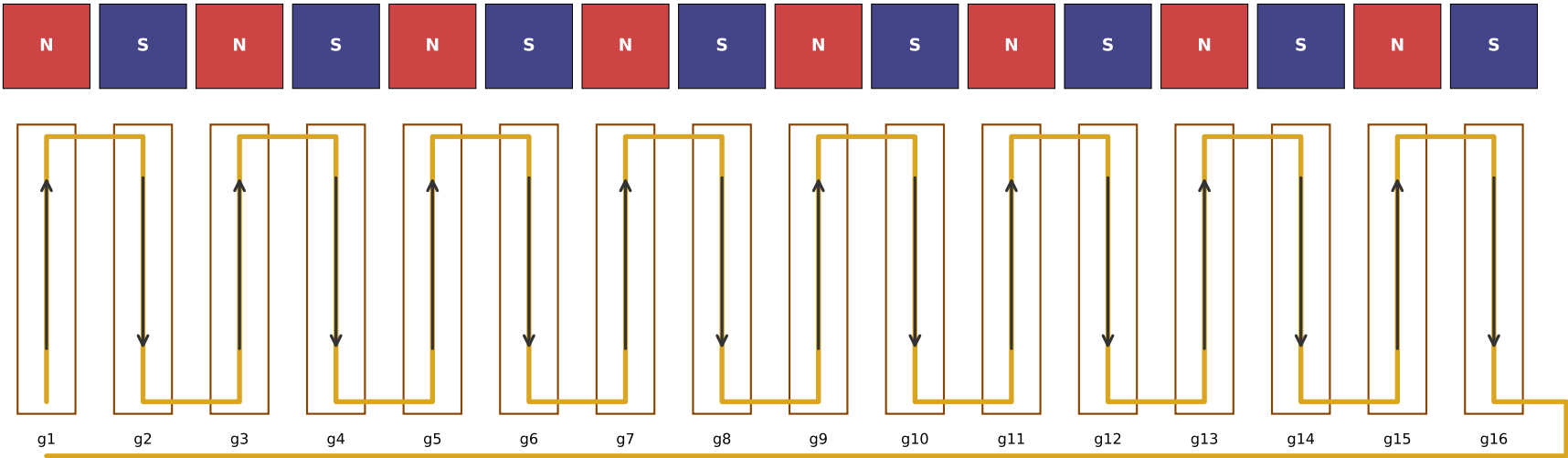
HF-RF16-SERP 16P/16S radial flux outer rotor, single-phase wave winding.



HF-RF16-SERP 16P/16S radial flux outer rotor, single-phase wave winding.

Unrolled 16-slot square-wave / serpentine winding

Magnets (rotor) — polarity above axial runs · serpentine pitch = pole pitch = 22.5°



26 AWG wave winding · target 80-120 turns · tails out cap · bridge + ≥1000 µF

Key dimensions (mm) — resolved geometry

Parameter	Value
MAG_COUNT	16
Pole/slot pitch	22.50°
Magnet face radius	56.38
Magnet back radius	61.38
Air gap	1.00
Sleeve outer / inner r	55.38 / 54.58
Bore diameter (radius)	94.00 (47.00)
Groove floor radius	49.00
Printed wall bore→floor	2.00
Copper radial fill	5.58
Groove / land arc @55.38	12.00 / 9.75
Rotor OD	132.00
Steel strip cut length	386

Geometry conflict resolution

CONFLICT: Original notes had drum bore r=47.00 with groove floor r=47.38, leaving only 0.38 mm printed wall (below the ≥2.00 mm minimum). Copper fill quoted as ~7.2 mm was inconsistent with a corrected sleeve outer r=55.38.

RESOLUTION (preferred):

- Magnet face r = 56.38, air gap = 1.00 exactly (≤1.2).
- Sleeve OUTER r = 55.38, wall = 0.80 → INNER r = 54.58.
- Bore RADIUS = 47.00 (diameter 94.00) for steel flux-return tube OD≈94.
- Groove floor r = 49.00 → printed wall = 2.00 mm.
- Copper radial fill = 55.38 – 0.80 – 49.00 = 5.58 mm (revises “7.2”).
- Groove open at r=55.38 before sleeve; land/groove arcs 9.75 / 12.00 at that radius sum to pitch arc 21.75 mm (16×22.5°) – no ±0.5 adjustment needed.
- Legacy BOM “steel tube OD~47” referred to the bore RADIUS figure; corrected to tube OD≈94 mm wall~3 length~28 to match Ø94 bore.

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Hardware BOM (bought) + key printed parts

Qty	Item	Notes
16	NdFeB Ø20×5	axially magnetized, N-S alt.
1	Steel strip 1.50×20.00×386	mild steel back-iron
1	Steel tube OD≈94 wall~3 L~28	flux return (bore Ø94)
2	608ZZ bearing	22×8×7
1	Shaft Ø8	through both bearings
—	M3 screws / heat-set inserts	rotor 6×; stator/cap 4×
—	M2 or snaps	sleeve clips
1 spool	26 AWG magnet wire	serpentine 80–120 turns
4	Diodes (bridge)	full-wave rectifier
1	Capacitor ≥1000 µF	DC filter
1	HF-RF16-ROTOR.stl	PETG/ABS/ASA/nylon
1	HF-RF16-STATOR.stl	grooves UP
1	HF-RF16-SLEEVE.stl	2×180° shells
1	HF-RF16-CAP.stl	bearing + grommets